

systemctl & journalctl Cheat Sheet

Start, stop and check services with systemctl; read their logs with journalctl.

Using it

```
systemctl COMMAND nginx ← ".service" can be left off · changing things needs sudo
journalctl -u nginx -f ← logs for one service, following new lines as they arrive
```

Controlling services

Command	Does	Command	Does
start / stop	Runs / stops it now	is-active	Prints active, inactive or failed
restart	Stops, then starts it	is-enabled	Prints enabled, disabled, masked...
reload	Re-reads config without stopping	is-failed	Exit code 0 if the service failed
status	State, PID, memory, last log lines	--failed	Lists every unit that has failed
enable / disable	Start / don't start it at boot	cat nginx	Shows the unit file and overrides
enable --now	Enable and start (disable --now)	edit nginx	Adds an override (see below)
mask / unmask	Blocks it from starting at all	daemon-reload	Re-reads unit files after edits

Reading systemctl status

```
* nginx.service - A high performance web server and a reverse proxy server
  Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
  Active: active (running) since Fri 2026-10-02 09:14:03 UTC; 1 day 2h ago
    Main PID: 815 (nginx) ← the main process
Oct 02 09:14:03 web1 systemd[1]: Started nginx.service - A high performance... ← last log lines
```

You see	It means	You see	It means
(path; ...)	The unit file actually in use	active (running)	Running normally
enabled	Starts at boot (static : on demand)	active (exited)	One-shot job, finished fine
preset: enabled	The distro's default setting	inactive (dead)	Stopped; nothing went wrong
masked	Blocked; unmask it first	failed	Crashed or exited with an error

journalctl (reading logs)

Option	Shows	Option	Shows
-u nginx	Only this service's logs	--since "1 hour ago"	From then on (or --until)
-f	Follows new lines live	-p err	Errors and worse only
-e	Jumps to the end of the log	-k	Kernel messages (like dmesg)
-n 50	Only the last 50 lines	--disk-usage	How much space logs use
-b	This boot (-b -1 : last boot)	--vacuum-time=2weeks	Deletes logs older than 2 weeks

Unit files and overrides

/etc/systemd/system/ (yours) wins over /usr/lib/systemd/system/ (packages: never edit, updates overwrite).

```
$ sudo systemctl edit nginx ← opens /etc/systemd/system/nginx.service.d/override.conf
[Service] ← add only the lines you want to change
Restart=on-failure ← save, then: sudo systemctl restart nginx
```

Handy recipes

Command	Does
journalctl -u nginx -b -p err	This service's errors since boot
journalctl -xeu nginx	Why did it just fail to start? (end of log, with hints)
sudo journalctl --vacuum-size=500M	Shrinks stored logs to 500 MB
systemctl list-unit-files --state=enabled	Everything set to start at boot

Remember: **enable** affects the next boot, **start** affects now; **enable --now** does both. After editing a unit file by hand, run **sudo systemctl daemon-reload** or systemd keeps using the old version.